RED SEAL EXAM PREP

421A Heavy Duty Equipment Technician

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

Pass mark: 70% · Aligned to the published occupational standard topic weightings

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Mock Exam — answer all questions, then check the key

Q1. A diesel engine exhaust has a continuous blue-grey smoke that does not clear after warm-up, regardless of load. Coolant level is stable and no coolant is in the oil. What is the MOST likely cause?

A) Engine oil is burning in the combustion chamber
B) Coolant leak into the combustion chamber — head gasket failure
C) Diesel fuel is contaminated with gasoline
D) Rich fuel mixture — injectors are over-fueling

Q2. A large loader tire has been assembled onto a multipiece rim and secured in an approved inflation cage. Under WorkSafeBC's OHS Regulation, what must the person inflating it do?

A) Set the regulator to the required pressure and step away
B) Stay clear while the beads seat, then approach to top up
C) Watch the lock ring closely while the pressure comes up
D) Stay clear of the trajectory and watch the pressure

Q3. A mine haul truck's cab pressure monitor reads below the required positive-pressure setpoint, but the pressurizer blower is running normally. What should the technician inspect first?

A) The blower motor brushes for arcing and high resistance
B) The recirculation damper spring for loss of tension
C) The condenser fan clutch for slippage under load
D) The fresh-air intake filter for dust loading and restriction

Q4. A hydraulic pump produces rated flow but system pressure builds slowly and takes longer than normal to reach the relief valve setting when a function is stalled. What is MOST likely?

A) Hydraulic fluid viscosity is too high — cold weather operation
B) Air in the hydraulic system from a suction leak
C) Main relief valve spring has weakened, lowering the cracking pressure
D) Pump displacement is reduced — high-pressure compensator is cutting in prematurely

Q5. While repairing a Weather Pack connector on a loader frame harness, a technician finds one unused cavity with nothing installed in it. What should be done with this cavity?

A) Leave it open so any condensation can drain freely
B) Fill it permanently with RTV silicone gasket maker
C) Pack it full of dielectric grease up to the shell face
D) Install a silicone cavity plug to seal out moisture

Q6. **A heavy equipment air brake system has a low air pressure warning light that activates at 80 psi but the system should have 120 psi. A compressor cycle test shows the compressor builds from 100 to 120 psi in 30 seconds and cuts out at 120 psi. What component is MOST likely leaking?**

A) Governor — cutting out prematurely at 120 psi

B) Air compressor — cannot maintain pressure after cut-out

C) Air dryer — moisture bypass is diluting the air supply

D) A system air leak downstream of the compressor

Q7. **In the correct order, the four strokes of a diesel engine cycle are:**

A) Intake, Compression, Power, Exhaust

B) Power, Intake, Compression, Exhaust

C) Intake, Power, Compression, Exhaust

D) Compression, Intake, Power, Exhaust

Q8. **A hydraulic oil sample is reported as ISO 4406 cleanliness code 22/20/17. What does this three-number code represent, and is this acceptable for a servo-hydraulic system with tight-tolerance components?**

A) The code indicates the oil has passed cleanliness testing — higher numbers indicate cleaner oil

B) Particle count ranges at 4µm, 6µm, and 14µm — heavily contaminated oil, not acceptable for servo systems

C) The three numbers represent viscosity, oxidation level, and water content — 22/20/17 is within normal limits for any hydraulic system

D) ISO 4406 only applies to transmission oils — hydraulic systems use a different cleanliness standard

Q9. **A technician reviews oil analysis results for a heavy equipment engine. The report shows: silicon = 28 ppm (normal <10), iron = 45 ppm (normal <30), lead = 8 ppm (normal <5). What do these results indicate together?**

A) Fuel dilution — diesel fuel contains trace silicon from fuel system wear

B) Coolant leak — high Si indicates antifreeze glycol contamination

C) Dirt ingestion causing accelerated iron wear and bearing wear

D) Normal wear patterns — all values within range for this hour interval

Q10. **A hydraulic system has a counterbalance valve installed on the rod side of a lift cylinder to prevent uncontrolled lowering under negative loads. The cylinder now lowers extremely slowly even with full spool travel. What adjustment is needed?**

A) Decrease the counterbalance valve setting — valve is set too high and is restricting free return flow

B) Increase the counterbalance valve setting — valve is holding at too high a pressure

C) Decrease the counterbalance valve pilot ratio — valve is not opening fast enough with available pilot pressure

D) Replace the counterbalance valve — slow lowering indicates internal failure

Q11. **To confirm that an alternator can still make its rated output, a technician clamps its output lead and switches on every accessory the machine carries. Why does that reading not prove the alternator's capacity?**

A) An alternator makes only the current the system draws

B) Output current must be read at the battery, not the alternator

C) The regulator holds output down once the battery is full

D) Accessory loads swing too much to give a steady reading

Q12. **A technician is diagnosing a hydraulic system with excessive heat generation. A thermal imaging scan shows the highest temperature at a specific control valve bank section. No external leakage is present. What is the MOST likely cause at that location?**

A) Internal spool leakage in that valve section

B) That section has a clogged work port filter

C) The port relief valve for that section is set too high

D) The section is working hardest — normal heat from useful work

Q13. **During an in-frame overhaul of a wet-sleeve diesel engine, the coolant side of the cylinder liners shows deep, clustered pitting. What caused this damage?**

A) Cavitation erosion from liner vibration and depleted coolant SCA

B) Electrolysis from stray electrical current in the cooling system

C) Scale deposits from hard water used to top up the coolant level

D) Acid etching from combustion gas leaking past the head gasket

Q14. **While reinstalling engine compartment sound insulation on a wheel loader, a technician finds several isolation mounts crushed flat and soaked with oil. What is the correct action?**

A) Leave them out, since the foam panels alone control the noise

B) Reuse them with new fasteners torqued tighter to take up the collapse

C) Replace the mounts, since a crushed mount transmits vibration

D) Shim behind them with washers to restore the original standoff

Q15. **A wheel loader with hydraulic-boosted service brakes has a spongy pedal that requires excessive travel to achieve normal braking. Fluid level is full and no external leaks are present. What is the MOST likely cause?**

A) Air entrained in the hydraulic brake system circuit

B) Brake disc/pads are worn beyond service limits

C) Hydraulic boost pressure is too low — booster assist is inadequate

D) Master cylinder primary cup has deteriorated — fluid bypasses the seal

Q16. **During routine maintenance, a technician measures the final drive planetary gear backlash on a large crawler dozer. The measurement is at the maximum specification limit. What does this indicate and what action should be taken?**

A) Backlash measurement is not meaningful for planetary gears — only ring gear condition matters

B) Significant final drive wear — document it, inspect the oil for debris, and plan replacement

C) Maximum backlash is preferred — it reduces gear-to-gear friction and heat generation

D) Maximum backlash requires immediate final drive disassembly and rebuild — the machine must not operate

Q17. **A dozer ripper penetrates poorly, and inspection shows the ripper tip worn far back with the shank protector missing entirely. If ripping continues in this condition, what is the most significant consequence?**

A) The ripper cylinder will cavitate under sustained down pressure

B) The tip retainer will friction-weld itself onto the shank

C) The beam pivot pins will spall from the added vibration

D) The shank itself wears away, forcing a costly shank replacement

Q18. **What is the primary function of a differential in a heavy equipment drive axle?**

A) To increase ground clearance

B) To increase the gear ratio for more pulling power

C) To allow drive wheels to turn at different speeds

D) To disengage drive power during braking

Q19. **Which type of hydraulic pump is most commonly used as the main pump in heavy equipment because of its high efficiency and ability to produce variable flow?**

A) Fixed-displacement external gear pump

B) Fixed-displacement vane pump

C) Variable-displacement axial piston pump

D) Internal gear (gerotor) pump

Q20. **What is the key operational difference between an open-center and a closed-center hydraulic system, and how does it affect fuel consumption?**

A) Open-center systems use hydraulic oil with an open (unsealed) reservoir; closed-center systems use a sealed pressurized reservoir — no difference in fuel consumption

B) Closed-center systems require a larger pump — the increased pump size consumes more fuel than an open-center design

C) Open-center systems circulate oil to tank in neutral; closed-center systems stop pump flow at standby, saving energy

D) Open-center is more efficient because continuous flow prevents pump cavitation, reducing overall energy use

Q21. **A diesel engine cranks but will NOT start. The fuel system has been verified. The NEXT most likely cause to check is:**

A) Faulty park brake switch

B) Alternator failure

C) Low engine compression

D) Low coolant temperature sensor

Q22. **A hydraulic accumulator has a pre-charge pressure of 900 PSI. The system minimum working pressure is 1,500 PSI and maximum is 3,000 PSI. The accumulator is found to be unresponsive — it discharges instantly with no sustained pressure release. What is the most likely cause?**

A) The accumulator check valve is stuck open, allowing immediate fluid return to tank

B) The bladder has burst — hydraulic fluid has entered the gas side and the accumulator is hydraulically locked

C) The pre-charge pressure is too high — 900 PSI exceeds system working pressure

D) The nitrogen pre-charge has leaked out, so the accumulator cannot store energy

Q23. **A technician is diagnosing an intermittent no-start condition. The engine cranks but does not start, and the problem occurs randomly. All fuel and compression tests are normal. A scan tool shows the engine speed sensor (MPU) has an intermittent signal. What is the MOST likely root cause?**

A) Chafed wiring or connector corrosion in the CKP sensor circuit

B) Engine speed sensor has failed internally — replace sensor

C) Sensor air gap is incorrect — engine vibration causes signal dropout

D) ECM is failing to process the CKP signal correctly

Q24. **The engine has started, but the starter motor keeps cranking after the key is released. What should the technician do immediately?**

A) Let it run - the overrunning clutch protects the starter drive

B) Rev the engine so the flywheel throws the pinion out of mesh

C) Disconnect the battery ground cable to stop the starter at once

D) Hold the key in the start position until the starter disengages

Q25. **A technician is working near a diesel engine exhaust system in an enclosed building. The PRIMARY hazard is:**

A) CO_2 (carbon dioxide) buildup causing fire risk

B) Particulate matter reducing visibility

C) NOx causing corrosion of tools

D) CO (carbon monoxide) exposure

Q26. **While rigging a transmission out of a grader during a scheduled service, a technician finds a sling with cut and frayed strands. Under RSOS A-2.04, what is required?**

A) Downrate it by half and finish this one lift

B) Tape the damaged area and complete the lift

C) Remove it from service, report and document it

D) Set it aside for the annual third-party inspection

Q27. **The Komatsu HB215LC hybrid excavator stores swing-regeneration energy in an ultracapacitor rather than a battery PRIMARILY because ultracapacitors:**

A) Charge and discharge much faster than batteries can

B) Operate below the 60 V DC high-voltage threshold

C) Store far more total energy for a given unit weight

D) Are unaffected by vibration and high ambient heat

Q28. **What is the purpose of the EGR (Exhaust Gas Recirculation) system on a modern diesel engine?**

A) To reduce NOx emissions by lowering peak combustion temperature

B) To preheat the intake air during cold starts

C) To improve fuel economy by recovering exhaust energy

D) To increase engine power by recirculating exhaust gases for a second combustion cycle

Q29. **After replacing a receiver-drier on a skidder, the technician pulls the system into a deep vacuum and holds it there before recharging. What is the main purpose of this evacuation step?**

A) To verify the clutch coil operates under low-load conditions

B) To boil off moisture and remove air from inside the system

C) To distribute refrigerant oil evenly through the system lines

D) To seat the compressor shaft seal before adding refrigerant oil

Q30. **A technician needs to replace a damaged section of 10-gauge wire in a circuit that carries 25 amps continuously. The only wire available is 14-gauge. Why is 14-gauge wire INAPPROPRIATE for this repair?**

A) The gauge number difference (14 vs 10) exceeds the maximum allowable splice ratio of 1.5 per circuit repair standard

B) 14-gauge wire is only rated for AC circuits — DC circuits require 10-gauge minimum

C) 14-gauge wire has a lower ampacity than 10-gauge and would overheat on a 25-amp circuit

D) 14-gauge wire has lower resistance than 10-gauge, which would increase current draw and damage the component

Q31. **A technician is lubricating all grease fittings on an articulated dump truck during a 250-hour service. One fitting accepts very little grease before pressure rises significantly. No grease appears at the joint seal. What should the technician do?**

A) Apply extra pressure — force the grease past the blockage

B) Investigate for a blocked fitting or a fully packed joint

C) Skip this fitting — the joint is adequately lubricated from last service

D) Replace the grease fitting — Zerk fittings fail over time

Q32. **A dozer at operating temperature cranks slowly. A clamp on the starter feed shows cranking draw well above the starter's specified figure. The starter has been removed and bench tested, and its free-running speed and current draw are within specification. What does this pattern point to?**

A) Batteries that cannot sustain the cranking load

B) Shorted windings inside the starter motor armature

C) Resistance in the starter feed and ground path

D) Mechanical drag in the engine or its accessories

Q33. **A technician is about to work with a cleaning solvent that has a GHS hazard pictogram showing a flame over a circle (oxidizer), a skull and crossbones, and an exclamation mark. What do these pictograms indicate and what PPE is REQUIRED?**

A) GHS pictograms are informational only — PPE selection is at the technician's discretion

B) These pictograms indicate the product is combustible (not flammable), mildly toxic, and irritating — standard work gloves and eye protection are sufficient

C) Oxidizer, acute toxicity, and irritant hazards — wear chemical-resistant gloves, eye/face protection, and apron per the SDS

D) Flammable, poisonous and corrosive hazards — flame-resistant coveralls and a respirator are required; glove choice is not critical

Q34. **A quick-release valve is installed close to the brake chambers. Its function is to:**

A) Limit maximum brake application pressure

B) Provide a secondary pressure source for emergency braking

C) Exhaust brake air rapidly at the chamber to reduce brake release time

D) Balance pressure between front and rear axle brakes

Q35. **A mounting bolt is missing from a dozer's certified ROPS. The shop has a grade 5 bolt of the same thread, length and head size on the shelf. What is the correct action?**

A) Fit the OEM-specified bolt at the OEM torque value

B) Fit the grade 5 bolt torqued to grade 5 specifications

C) Fit the grade 5 bolt now and install the OEM bolt later

D) Fit the grade 5 bolt with a lock washer and thread locker

Q36. **What does the lock-up clutch in a torque converter do once it engages?**

A) Couples the engine to the turbine mechanically

B) Lets the stator freewheel at high turbine speed

C) Multiplies engine torque at the turbine output

D) Holds the converter charged with cooled oil

Q37. **A machine's CAN bus (Controller Area Network) has a fault code indicating loss of communication with the transmission control module (TCM). The TCM powers up normally when tested independently. What is the MOST likely cause?**

A) TCM internal control circuitry has failed

B) Termination resistor at the TCM is defective

C) A wiring fault in the CAN H/CAN L pair at the TCM node

D) ECM has failed — it controls all CAN communication

Q38. **A technician uses oil analysis to monitor a diesel engine. The report shows TAN (Total Acid Number) of 4.0 mg KOH/g compared to a new oil baseline of 1.2 mg KOH/g. The oil is at 75% of change interval. What should the technician do?**

A) Change oil immediately — elevated TAN indicates acid buildup that causes corrosive wear

B) Continue to the full change interval — TAN of 4.0 is within normal range

C) Add an oil neutralizer additive to reduce TAN

D) Increase the change interval — high TAN indicates the oil is still absorbing acids effectively

Q39. **After rebuilding a hydraulic brake caliper, the brake drags during initial operation. The caliper pistons were replaced and the slides were lubricated. During diagnosis, the brake hose is found to be deteriorated internally. How does a deteriorated brake hose cause brake drag?**

A) A deteriorated hose restricts brake fluid flow, reducing hydraulic pressure to the caliper and causing partial engagement

B) A deteriorated hose allows air ingestion into the system, creating a partial air lock that holds the caliper pistons applied

C) The collapsed inner lining acts as a one-way check valve — fluid enters under pedal pressure but cannot return

D) Internal hose deterioration reduces hose flexibility, causing it to transmit vibration that continuously applies the brake pedal

Q40. **On a hybrid excavator, which of the following qualifies as an 'insulated tool' of the kind CSA Z462 requires where contact with energized parts is possible?**

A) A tool marked with the double-triangle 1000 V symbol

B) A wrench with moulded rubber grips from the shop cart

C) A chrome wrench wrapped in electrical tape by the tech

D) A tool with a plastic-coated handle for corrosion control

Q41. **After installing a wheel on an articulated dump truck, the technician torques the nuts to specification in a star (criss-cross) sequence. The required follow-up is to:**

A) Coat each stud with thread-locking compound while warm

B) Loosen and fully re-seat the wheel after the first shift

C) Re-check the nuts only at the next 250-hour service

D) Re-torque the nuts after the first short period of operation

Q42. **A differential lock is engaged on a machine stuck in mud. Once the machine is free, the operator MUST:**

A) Rev the engine to seat the lock before releasing

B) Disengage the diff lock before turning

C) Shift to a higher gear immediately

D) Leave it engaged for better traction on the road

Q43. **A turbocharged diesel engine exhibits excessive black smoke at light loads and hesitation when the throttle is quickly applied. Under hard acceleration the smoke clears and power is adequate. What is the most likely cause?**

A) EGR valve stuck closed — recirculation gases are being replaced by excess raw fuel

B) Turbocharger response lag — insufficient boost at low RPM causes a rich condition until boost builds

C) Injectors are worn and delivering excess fuel at all loads — full power at WOT confirms the pump is adequate

D) Fuel injection timing is advanced — early injection causes smoke at light loads but clears under load

Q44. **A dozer equipped with a certified ROPS was involved in a rollover, and the ROPS posts are visibly bent. What must happen before the machine returns to service?**

A) Straighten the posts using controlled heat and jacks

B) Reinforce the bent areas with welded fish plates

C) Replace the ROPS per the manufacturer's requirements

D) Sleeve the bent posts with heavy-wall structural tubing

Q45. **A diesel engine produces black smoke under full load but white smoke during warm-up that clears when hot. The fuel system and turbo are confirmed serviceable. What is the MOST likely cause of the black smoke under full load?**

A) Air filter restriction — insufficient air for complete combustion at full load

B) Engine is operating normally — black smoke at full load is acceptable for older diesels

C) Injector timing advanced — fuel injected too early causing incomplete combustion

D) Coolant leak into combustion chamber — white smoke becomes black as load increases

Q46. **In a planetary gear set where the RING GEAR is the input, the PLANET CARRIER is held (stationary), and the SUN GEAR is the output — what type of motion results?**

A) Speed increase (overdrive) with rotation reversal

B) Speed reduction with same direction rotation

C) The system locks up — no output is possible with carrier held

D) 1:1 ratio — no gear change

Q47. **The 421A Red Seal Occupational Standard defines the welding an HDET performs under A-2.05 as basic welding. Which description matches the scope the standard sets?**

A) Structural welding once a shop supervisor signs off on it

B) Non-structural, non-pressure oxy-fuel and arc welding

C) Any arc welding on the machine except on pressure vessels

D) All welding on the machine's own frame, booms and buckets

Q48. **A machine with a hydrostatic drive system loses tractive effort in both forward and reverse but the engine power appears normal. The MOST likely cause is:**

A) The variable displacement pump swashplate is not responding to operator input — stuck at minimum displacement

B) The travel motor has seized — blocking flow in the circuit

C) Low charge pump pressure causing the hydrostatic loop to cavitate

D) Transmission oil temperature is too low, causing high viscosity

Q49. **A technician measures 3,200 psi system pressure during a stall test (actuator held against load) but only 1,800 psi during normal operation. The main relief is set to 3,500 psi. What does this indicate?**

A) Normal operation for a load-sensing pressure-compensated system

B) Pump is worn and cannot build full pressure under flow demand

C) System is cavitating — install a boost pump

D) Main relief valve is defective — it should hold constant pressure

Q50. **During hybrid-systems training on a Cat 336E H excavator, an apprentice asks where the machine stores the energy it recovers while braking the upper structure swing. The correct answer is:**

A) In a chassis-mounted ultracapacitor module

B) In a motor-driven steel flywheel assembly

C) In nitrogen-charged hydraulic accumulators

D) In a lithium-ion traction battery package

Answer Key & Explanations

Q1. Answer: A

Blue-grey continuous smoke = oil burning. Unlike white smoke (coolant/unburnt fuel) or black smoke (excess fuel), blue/grey indicates engine oil is entering the combustion chamber. Check: 1) Oil level and consumption rate. 2) Valve stem seals (oil drips into combustion chamber, worse at startup/deceleration). 3) Piston rings and cylinder wall wear. 4) Turbocharger seal (oil drawn into intake manifold). Stable coolant eliminates head gasket.

Q2. Answer: D

Inflation is done from outside the line the rim parts would take if the assembly let go, and the gauge is watched the whole time. WorkSafeBC OHS Regulation section 16.16(4) reads: "A person inflating a tire must stay out of the potential explosion trajectory, if it is practical to do so, and actively monitor the tire's inflation pressure." Both halves carry weight. Standing where the lock ring can be watched puts the technician in the trajectory, which is exactly where the parts travel, and it is a position workers have been killed in. Setting a regulator and walking away leaves nothing monitoring the pressure, and an assembly taken past its pressure is how components get launched. Approaching once the beads have seated puts a body back in the trajectory for the rest of the fill. The cage answers a separate requirement — section 16.16(7) does not allow a multipiece rim to be inflated to operating pressure until it is mounted onto the mobile equipment or secured in a cage or engineered containment device — but a restraining device does not remove the duty to stay clear and watch the gauge.

Q3. Answer: D

A dust-loaded fresh-air filter is the most common cause of lost cab pressure. As the intake filter loads up, airflow into the cab drops until the system can no longer push in more air than leaks out, and cab pressure falls even though the blower runs normally. Filters on machines in heavy dust must be changed based on restriction and pressure readings, not on calendar intervals.

Q4. Answer: B

Air in hydraulic oil compresses (oil is incompressible but air is not), causing slow/spongy pressure buildup. Air can enter from a suction side leak (low-pressure side of pump), a loose fitting, or aeration from a low fluid level. Symptoms: slow pressure buildup, foamy or aerated oil in reservoir, noisy pump (cavitation), sluggish actuators. Fix: find and eliminate the air entry point, purge system, check fluid level.

Q5. Answer: D

Every unused Weather Pack cavity must be filled with a cavity plug — a sealed connector only works if every cavity is sealed. One open cavity lets moisture and dust into the whole connector shell, causing green terminal corrosion, high resistance, and intermittent faults on the wired circuits beside it. The system uses self-lubricating silicone cable seals crimped to each terminal and matching plugs for empty positions. Grease and RTV are not substitutes: they wash out or block future service, and neither seals the cavity interface reliably.

Q6. Answer: D

Compressor builds to 120 psi correctly, but system drops to 80 psi triggering the warning — this is a downstream leak, with air escaping faster than the compressor maintains when stationary. The compressor output is confirmed good (builds pressure, cuts out at spec). The pressure drop after cut-out is caused by air escaping through a leak in the system (brake chambers, supply lines, fittings, drain valves). Perform a static leak test: engine off, brakes released, monitor pressure drop rate. Should not exceed 2 psi/minute for service brakes.

Q7. Answer: A

Intake → Compression → Power → Exhaust. During intake, air fills the cylinder. Compression heats the air to ~500°C+. Diesel injected at TDC auto-ignites (power stroke). Exhaust gases are expelled. Diesel engines have no spark plugs — compression alone ignites the fuel.

Q8. Answer: B

ISO 4406: three numbers = particle count ranges at ≥4µm, ≥6µm, ≥14µm. Higher number = dirtier oil. Each increment of 1 in the ISO code represents doubling the particle count. Code 22 means 20,000–40,000 particles per mL at ≥4µm. Servo valves have bore clearances of just 1–5µm — even particles much smaller than visible can cause valve erosion and stiction, so servo systems typically require ISO 16/14/11 or cleaner. Typical cleanliness requirements: servo valves = ISO 16/14/11; piston pumps = ISO 17/15/12; gear pumps = ISO 20/18/15. Code 22/20/17 would rapidly destroy precision components.

Q9. Answer: C

High silicon = dirt contamination. High iron = accelerated component wear (contamination-induced). High lead = bearing wear. These three together tell a clear story: dirt entered through the air intake (or oil fill), caused abrasive wear on iron components, and the abrasive material has reached and damaged lead-alloyed bearings. Immediate actions: 1) Change oil and filter immediately. 2) Investigate air filtration — find and repair the air filter, breather, or cover sealing problem. 3) Check air filter condition and sealing.

Q10. Answer: A

A counterbalance valve set too high requires higher pilot pressure to open — restricting lowering speed. Counterbalance valve setting should be 1.3× the maximum load-induced pressure. If set above this, the valve cracking pressure requires excessive pilot pressure to open. Lower the setting to allow adequate pilot-to-open pressure at normal operating conditions, while still preventing uncontrolled drop.

Q11. Answer: A

An alternator supplies the current the loads and the battery ask for, and no more, so a reading taken under the machine's own accessories measures the load, not the alternator. On most machines the accessories together add up to well under the alternator's rating, so a healthy unit reads far below its rating on the clamp and the technician who reads that as a fail condemns a good alternator. Measuring capacity means asking the system for at least the rated current: connect a carbon pile across the battery posts, hold the engine at the speed the test procedure specifies, and increase the load until system voltage is pulled down to the value that procedure names, then read the clamp on the output lead and compare it with the rating. Carbon piles heat quickly, so the load is applied only briefly. Where the design allows it, full-fielding the alternator does the same job from the other direction by taking the regulator out of the loop. The regulator is not what spoils the test - it holds its set voltage whatever the battery's state of charge and does not throttle an alternator back because a battery is full. Switching every accessory on does give a steady enough reading; the trouble is the size of the load, not its steadiness. And output is read at the alternator's own output lead for a reason: a clamp at the battery shows only the net charge or discharge left over after the machine's loads have taken their share.

Q12. Answer: A

A hot control valve section without external leakage indicates internal spool leakage converting high-pressure flow to heat. Fluid bypasses internally across the spool lands from high pressure to tank at a pressure drop — converting hydraulic energy to heat at that location. A relief valve set too high never cracks, so it passes no oil and makes no heat at all. Test: isolate that section and compare system heat generation. If heat drops, the valve section is the source. Rebuild or replace the section with worn spool clearances.

Q13. Answer: A

Coolant-side liner pitting = cavitation erosion — prevented by maintaining nitrite-based SCA levels. Each combustion event makes the wet sleeve vibrate like a bell. As the liner wall snaps away from the coolant, vapour bubbles form; when it snaps back, the bubbles implode against the surface and hammer out pits that can eventually perforate into the cylinder. Nitrite in the supplemental coolant additive (SCA) package forms a sacrificial oxide film that absorbs the implosions. Test coolant with SCA strips at PMs and maintain roughly 1.5–3.0 units per gallon, or use an extended-life coolant per OEM.

Q14. Answer: C

Sound suppression on heavy equipment works two different ways at once, and the two are not interchangeable. Foam and insulation panels absorb airborne noise, while isolation mounts break the structural path by letting an elastomer element deflect so vibration is not carried into the frame and up into the operator station. Once the elastomer is crushed flat or has swollen and softened from oil contamination it no longer deflects, so it is a solid metal-to-metal path and structure-borne noise and vibration go straight through it. No quantity of absorptive foam corrects that, because it is not an airborne problem. The occupational standard names foam, insulation, panels, fasteners and mounts together as the sound suppression components precisely because a complete repair addresses both paths, and noise and vibration are the named symptoms.

Q15. Answer: A

Air in a hydraulic brake circuit is compressible — unlike brake fluid — causing spongy, high-travel pedal. Air can enter through: micro-leaks in connections (suction side), air bubbles from aeration (low fluid level at some point), or improper bleeding procedure. Systematic bleeding from the furthest wheel cylinder to the nearest purges air. If bleeding does not correct, suspect a faulty master cylinder seal allowing air to be drawn back in.

Q16. Answer: B

Backlash at maximum spec = advanced gear and bearing wear — document the finding, inspect the oil for metallic debris, and schedule replacement. Backlash is the free play between mating gear teeth. New gears have minimum backlash; as gear flanks wear, backlash increases. Maximum specification is the engineering limit before tooth contact geometry becomes unsafe. At maximum, while technically still within specification and legal to operate, the wear rate accelerates (less tooth engagement area), oil analysis will show elevated iron/chromium particles, and the next inspection may find it over-limit. Scheduling final drive replacement before the next major interval prevents catastrophic failure and unplanned downtime in the field. Planetary final drives on dozers are expensive to replace in the field.

Q17. Answer: D

Tips and shank protectors are sacrificial parts that shield the expensive shank. A worn-back tip and missing protector expose the shank leading edge directly to rock abrasion. The shank is a major structural component, so letting it wear costs far more than replacing the tip and protector on schedule. Poor penetration is the operational symptom that the wear parts are past their service point.

Q18. Answer: C

Differential allows speed difference between drive wheels when turning. On a turn, the outer wheel travels farther than the inner wheel and must rotate faster. Without a differential, tires would scrub and drivetrain stress would be extreme. The differential uses bevel gears to split torque while allowing different wheel speeds.

Q19. Answer: C

Axial piston pumps are the industry standard for heavy equipment. They offer high efficiency, high pressure capability, and can be built with variable displacement (servo-controlled swashplate). Gear pumps are simpler but fixed displacement. Vane pumps are less common in heavy equipment.

Q20. Answer: C

Open-center: continuous flow through the valve centers back to tank at low pressure when valves are in neutral (pump always spins fluid). Closed-center: pump output stops (variable displacement) or pressure is held at standby when no function is active — more energy efficient at idle. Open-center control valves allow oil to flow from pump → through the valve center → back to tank when in neutral. The pump always delivers flow (fixed displacement), consuming engine power. Closed-center valves block flow in neutral — with a variable displacement pump, the pump destrokes to minimum displacement (low energy use). This is why modern high-end equipment uses closed-center with load-sensing systems.

Q21. Answer: C

No-start with verified fuel = check compression. Diesel ignition requires sufficient compression heat (~450°C minimum) — confirm with a compression or leak-down test. Worn rings, burnt valves, or a blown head gasket can reduce compression below ignition threshold. Cranking speed also affects compression — check battery and starter too.

Q22. Answer: D

Accumulator discharges instantly = lost nitrogen pre-charge. Accumulators store energy by compressing a nitrogen pre-charge with hydraulic fluid. Without pre-charge (bladder collapsed against the outlet), there is no spring force to push fluid out — it releases immediately and with no stored energy. Check: isolate the accumulator from the system and measure gas port pressure with a nitrogen pressure gauge. Pre-charge should be approximately 60–90% of minimum working pressure (900 PSI against 1,500 min is within typical range).

Q23. Answer: A

Intermittent faults are almost always wiring/connector issues causing an intermittent open circuit, not component failures. A failed sensor typically produces a continuous fault — it works or it doesn't. An intermittent signal that changes with vibration, temperature, or movement indicates a chafed wire, corroded connector pin, or broken wire strand that makes intermittent contact. Check the harness for damage near moving components and clean/reseat the connector.

Q24. Answer: C

Open the circuit first, diagnose second. A starter that keeps cranking after the key is released has a circuit that is still made: welded solenoid contacts, a stuck start relay, or an ignition switch that has not returned. The overrunning clutch stops the engine from driving the armature, but it does not stop the motor from running on its own supply, and a starter running unloaded at full speed overheats and throws its windings in seconds while the ring gear and battery pay for the delay. Pulling the ground cable opens the whole system with one connection. Then find why the circuit stayed closed - welded contacts weld again and are replaced, not filed.

Q25. Answer: D

Carbon monoxide (CO) is the primary threat. CO is colourless, odourless, and immediately dangerous to life — it binds to hemoglobin 240x more effectively than oxygen, and you cannot smell or see it. Symptoms appear rapidly in enclosed spaces. Ensure adequate ventilation or use CO monitors whenever running engines indoors.

Q26. Answer: C

A-2.04 splits this into three linked performance criteria. The technician inspects the equipment under .12P, repairs or replaces defective gear and reports it under .13P, and documents the maintenance information under .14P. All three are required, because removing a bad sling from the rack protects the next lift while the report and the record protect everyone who comes after. A-2.04.04P adds that rated capacities are determined by referring to tags and specifications, which is the reason downrating by a guessed factor is not an option: nobody can calculate the residual strength of a cut sling from the outside.

Q27. Answer: A

Ultracapacitors excel at power density — rapid charge and discharge. Excavator swing cycles repeat every few seconds, so the storage device must absorb braking energy and release it for acceleration almost instantly. Capacitors store energy electrostatically (ion migration) rather than through chemical reactions, allowing much faster power transfer than a battery. Their weakness is low total energy density, which is why they suit short, frequent cycles rather than long-duration storage.

Q28. Answer: A

EGR: introduces cooled exhaust gas into the intake, lowering combustion temp = lower NOx. NOx (nitrogen oxides) forms at extremely high combustion temperatures. Recirculating inert exhaust gas into the intake dilutes the oxygen content and absorbs heat, reducing peak combustion temperature and NOx formation. EGR is part of the EPA Tier 4/Euro 6 emissions compliance strategy along with SCR/DEF.

Q29. Answer: B

Evacuation removes air and boils trapped moisture out of the system. Under deep vacuum, water's boiling point drops low enough that moisture vaporizes and is pumped out; holding the vacuum also confirms the system is leak-tight. Moisture left behind forms corrosive acids with refrigerant and oil and can freeze at the metering device, while residual air acts as a non-condensable gas that drives head pressure up.

Q30. Answer: C

Wire ampacity: lower gauge number = thicker wire = higher ampacity. AWG 14-gauge is rated for approximately 15 amps maximum in chassis wiring. A 25-amp circuit exceeds this by 67%. Undersized wire develops resistance heating proportional to I^2R — at 25A through 14-gauge, the wire will overheat, melt and damage insulation, and create a fire and circuit failure hazard. The original 10-gauge wire is rated for approximately 30 amps continuous. Always match or exceed the original wire gauge in repairs. Approved repair: use 10-gauge or 8-gauge (heavier) wire.

Q31. Answer: B

A grease fitting that builds pressure immediately without taking grease indicates a blockage — do not overpressure; investigate why grease is not moving. Possible causes: 1) Fitting check ball is stuck (clean or replace fitting). 2) Grease passage in the pin is blocked with hardened old grease. 3) The joint is completely packed (correct — no more grease needed). 4) The pin is seized. Applying excessive force can damage seals. Diagnose before applying pressure: remove the fitting and verify it passes air.

Q32. Answer: D

A starter draws current in proportion to the load it turns, so draw above specification with the crank speed down means the motor is being loaded, not starved. Put extra resistance anywhere in the feed or the return path and less voltage reaches the motor: it draws less current than its specification, makes less torque, and cranks slowly. A battery down on charge or capacity produces the same pattern for the same reason. Both of those faults read below the specified draw, which is the opposite of what the clamp shows here, and both are chased down with a voltage-drop test on each side of the circuit and a battery load test. Draw above specification is the other half of the picture: something is making the armature work harder than it should, so it turns more slowly, and as the speed falls the counter-voltage the armature generates falls with it and lets still more current through. That load sits either inside the starter or outside it, and the bench test separates the two, because a starter with shorted armature windings shows excess draw while running free on the bench and this one ran to specification. What is left is drag on the engine side: a failing main or rod bearing, a driven accessory such as a hydraulic pump or air compressor beginning to seize, or an internal failure that has started to bind. Bar the engine over by hand with the starter out, and turn each driven accessory separately, before any parts are ordered.

Q33. Answer: C

GHS pictograms: flame-over-circle = oxidizer (can intensify fire), skull/crossbones = acute toxicity (category 1–3, may be fatal), exclamation mark = harmful/irritant/sensitizer. WHMIS 2015 (aligned with GHS) uses standardized hazard pictograms on all hazardous workplace products. A plain flame means flammable; a flame over a circle means oxidizer — it need not burn itself, but it can violently intensify a fire or explosion when combined with flammables, so strict storage segregation is required. Skull/crossbones: acute toxicity data indicates fatal or serious effects — do NOT use without consulting the SDS and following respiratory and skin protection requirements. Required PPE must match the SDS Section 8 recommendations — chemical-resistant gloves, face shield or goggles, a chemical-resistant apron, and potentially a supplied-air respirator depending on exposure limits — not the technician's preference.

Q34. Answer: C

Quick-release valve exhausts air locally at the brake chamber. When the driver releases the brake pedal, air must exhaust. Without a quick-release valve, air would have to travel all the way back to the foot valve to exhaust — slow release means slow brake release (brake drag). The quick-release valve vents air at the chamber immediately.

Q35. Answer: A

ROPS mounting hardware is part of the certified structure, so only the specified fastener at the specified torque is acceptable. The mounting bolts are what transfer rollover loads from the structure into the frame, and their grade, size and clamping force are established by the certification, not chosen in the shop. Substituting a lower grade changes the joint's clamp load and yield behaviour, and fitting non-specified hardware is a modification of a certified structure, which under provincial OH&S must be done in accordance with the manufacturer's instructions and re-certified as restored to its original performance requirements by the equipment manufacturer or a professional engineer.

Q36. Answer: A

The lock-up clutch takes the fluid out of the drive path. A torque converter always slips a little, and slip becomes heat and burnt fuel. When conditions allow it - enough speed, light enough load, oil warm - the transmission control applies a clutch that locks the converter cover to the turbine, giving a solid connection with no slip at all. Torque multiplication is the stator's work and is over long before that point, and the stator freewheels in the coupling phase whether the clutch is applied or not. A worn or cycling lock-up clutch shows up as shudder on apply, high converter-out oil temperature and lost fuel economy.

Q37. Answer: C

TCM works independently but not on CAN = communication wiring fault (open or short in the twisted pair between the TCM node and the bus backbone), not TCM failure. CAN bus uses twisted pair (CAN H and CAN L). Faults: broken wire (open), shorted wires, damaged shield, missing termination resistor. Test: measure CAN bus resistance with ignition off (should be ~60 Ω for a properly terminated bus). Incorrect resistance = missing/failed termination resistor. Use a CAN bus analyzer or oscilloscope to find break location.

Q38. Answer: A

High TAN indicates acid accumulation in the oil from combustion byproducts and oil oxidation. New oil TAN: ~1.2 mg KOH/g. A TAN of 4.0 mg KOH/g at 75% interval indicates the oil has lost its alkalinity reserve (TBN) and is now acidic. Acidic oil causes corrosive wear on engine bearings and cylinder walls. Change oil immediately and investigate root cause (overextended intervals, blowby contamination, fuel dilution).

Q39. Answer: C

Deteriorated brake hose: collapsed inner lining acts as a check valve — fluid goes in but can't return, so the pistons remain applied after the pedal is released. The outer appearance of a brake hose can look fine while the inner lining delaminates and collapses. Under pedal application, pressure pushes past the blockage and applies the brakes. When the pedal is released, the return path is blocked by the collapsed lining — the caliper remains applied. Test: with brakes applied, crack the caliper bleeder valve — if the piston releases immediately when bleeder is opened, the restriction is in the hose.

Q40. Answer: A

An insulated tool is manufactured and individually dielectrically tested, then marked with the double-triangle symbol and its voltage rating, commonly 1000 V. The marking is the evidence that the tool was proof tested; without it, nothing about a handle covering has been verified. Comfort grips, corrosion coatings and wraps of electrical tape are insulating in appearance only: they are not continuous over the shank, they were never tested, and tape unwinds under torque. Insulated tools must also be inspected before every use, because a nick, crack or embedded metal chip in the coating destroys the protection.

Q41. Answer: D

New wheel installations must be re-torqued after a short break-in period. As the mating surfaces of the hub, wheel, and nuts seat under load and thermal cycling, clamping force can relax below spec, which is how wheel-off incidents start. Manufacturers typically call for a re-torque within roughly the first 10 service hours or 50 to 100 km of operation, then regular checks afterward. The star sequence ensures the wheel seats evenly and concentrically at every torque pass.

Q42. Answer: B

Diff lock must be disengaged before turning or driving at normal speeds. Diff lock forces both wheels to rotate at identical speed. During a turn, the outer wheel needs to rotate faster — with diff lock engaged, tires scrub, drivetrain is stressed, and steering becomes very difficult. Only use on straight, low-traction surfaces.

Q43. Answer: B

Turbo lag: black smoke at light load, clears under hard acceleration. At low loads and RPM, the turbo hasn't spun up to produce adequate boost. The engine receives excess fuel relative to available air — resulting in a rich/incomplete combustion (black smoke). As load increases, exhaust energy spins up the turbocharger, boost pressure rises, and the air-fuel ratio normalizes — the smoke clears. This is normal behaviour but excessive lag can indicate a worn or damaged turbocharger.

Q44. Answer: C

A ROPS that has absorbed a rollover is spent and must be replaced or repaired only under manufacturer certification. The structure protects by controlled deformation, so bent members have already used their energy-absorbing capacity. Heating, straightening, sleeving, or plating alters the certified material properties and voids the certification. Only the manufacturer or an authorized engineer can recertify a protective structure.

Q45. Answer: A

Black smoke = rich mixture (excess fuel, insufficient air). White smoke during warm-up is normal (unburnt fuel/water condensation) and clears when hot. Black smoke at full load with serviceable fuel system points to air starvation — the air filter is restricting intake. The engine can only burn as much fuel as available air allows. Check air filter restriction indicator; replace if above spec.

Q46. Answer: A

Ring input + carrier held + sun output = speed increase with a change of direction. With the carrier held, the planets turn on fixed centres and act as idlers: the ring drives them, they drive the sun, and the sun turns the opposite way. The sun makes Nring ÷ Nsun turns for every turn of the ring, so a 72-tooth ring driving an 18-tooth sun spins the sun four times faster and backwards. The 1 + (Nring ÷ Nsun) formula belongs to the held-ring case with the carrier as output, and does not apply here.

Q47. Answer: B

The standard caps the HDET at non-structural, non-pressure oxy-fuel and arc welding. That boundary is deliberate and it explains why sub-task H-37.03 is titled Performs mechanical repairs on structural components rather than repairs: bracket tabs, guards and non-load-path items are in scope, while frame rails, cross-members and protective structures are not. Knowing where your own ticket stops is a tested competency, because welding outside it can void a certification and shift liability onto you and the shop.

Q48. Answer: C

Low charge pressure = hydrostatic drive weakness in both directions. The charge pump maintains circuit pre-charge pressure (typically 200–400 PSI) to prevent cavitation and replenish leakage. If charge pressure is low, the main pump cannot develop full displacement in either direction. Symptoms: weak in both forward and reverse, no fault codes on simpler systems. Check charge pressure with a gauge on the charge port.

Q49. Answer: A

Load-sensing (LS) pressure-compensated systems are designed to operate at load-demanded pressure, not maximum — pressure is reduced proportional to demand. At full stall (no flow needed), the pump builds to relief setting minus LS differential (typically 200-300 psi margin). At normal operation with flow, pump pressure matches load requirement. 1,800 psi during operation means the actual load demands 1,800 psi — perfectly normal for load-sensing systems.

Q50. Answer: C

The Cat 336E H is a hydraulic hybrid — swing energy goes into nitrogen gas accumulators. Swing-brake energy compresses nitrogen in accumulators and is released to help accelerate the next swing, managed by the ESP pump and ACS valve. This contrasts with the Komatsu HB215LC, which uses an electric swing motor-generator and an ultracapacitor. Knowing which hybrid architecture a machine uses determines whether HV electrical precautions even apply.